

NIZHYN, Ukraine

WMO: 332460

Lat: 51.0381N

Lon: 31.9003E

Elev: 126

StdP: 99.82

Time Zone: 2.00 (EUE)

Period: 94-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.9	-16.1	-21.6	0.6	-18.0	-19.2	0.7	-15.7	7.2	-1.7	6.4	-2.7	1.3	40	0.476

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	30.1	20.7	28.5	19.9	27.1	19.2	22.1	28.0	20.9	26.8	20.0	25.5	1.7	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.8	14.7	25.6	18.8	13.8	24.1	17.9	13.1	23.0	65.1	28.0	61.1	27.0	57.9	25.5	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.2	5.3	4.5	-22.6	32.7	3.7	1.7	-25.2	33.9	-27.4	34.9	-29.5	35.9	-32.2	37.1		
(5)			-22.8	23.8	3.7	1.8	-25.4	25.2	-27.6	26.2	-29.7	27.3	-32.4	28.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.8	-4.6	-2.5	1.7	9.0	14.8	18.0	20.2	18.4	13.2	7.1	1.9	-4.1		
	DBStd	9.88	5.63	5.42	4.31	4.45	4.29	3.40	2.80	3.12	3.93	4.38	5.03	6.01		
	HDD10.0	1948	453	349	258	71	9	0	0	0	14	112	245	437		
	HDD18.3	4012	711	582	516	281	124	48	13	38	159	349	494	696		
	CDD10.0	1152	0	0	1	40	158	241	317	261	111	21	1	0		
	CDD18.3	173	0	0	0	1	16	39	72	41	6	0	0	0		
	CDH23.3	1649	0	0	0	3	152	337	637	445	72	3	0	0		
	CDH26.7	400	0	0	0	0	22	69	177	119	14	0	0	0		
(14) Wind	WSAvg	1.9	2.2	2.3	2.4	2.0	1.8	1.7	1.4	1.3	1.5	1.7	2.1	2.2		
(15) Precipitation	PrecAvg	609	37	34	35	47	74	66	73	56	58	46	49	37		
	PrecMax	747	74	61	76	110	148	100	194	124	125	124	122	77		
	PrecMin	392	12	9	3	8	35	24	19	1	8	3	3	9		
	PrecStd	96	14	16	19	27	33	26	46	35	36	31	28	18		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.7	8.4	16.1	23.4	28.6	30.3	32.2	32.2	28.3	22.5	13.7	7.7		
		MCWB	4.9	6.1	8.8	14.7	18.2	20.5	22.0	21.0	18.2	16.5	11.1	6.3		
	2%	DB	4.2	6.3	12.8	21.0	27.0	28.7	30.3	29.8	25.4	18.7	11.4	5.9		
		MCWB	3.0	4.6	7.6	13.1	18.0	19.9	21.2	20.4	17.1	14.2	9.7	5.1		
	5%	DB	2.8	4.8	10.3	18.8	25.1	27.1	28.7	27.6	22.7	15.7	9.8	4.4		
		MCWB	1.8	3.4	6.3	12.1	17.7	19.3	20.5	19.2	15.9	12.6	8.7	3.7		
	10%	DB	1.7	3.5	8.0	16.8	22.9	25.2	27.1	25.7	20.1	13.6	8.3	2.6		
		MCWB	0.9	2.5	4.7	11.1	16.2	18.4	19.9	18.1	14.8	11.1	7.2	1.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.1	6.8	10.0	15.7	20.8	22.2	23.2	22.9	19.5	17.3	11.7	6.7		
		MCDB	5.4	8.1	15.0	21.5	25.2	27.9	29.7	29.8	26.7	21.3	13.4	7.3		
	2%	WB	3.1	4.9	8.2	14.1	19.2	21.1	22.4	21.1	18.0	14.7	10.0	5.1		
		MCDB	3.8	6.0	12.1	19.1	24.8	27.0	28.6	27.7	23.5	17.7	10.9	5.8		
	5%	WB	1.9	3.7	6.7	12.9	18.0	20.1	21.4	20.0	16.7	13.0	8.8	3.8		
		MCDB	2.6	4.7	9.7	17.2	23.0	25.6	27.2	25.9	21.3	15.6	9.8	4.4		
	10%	WB	1.0	2.6	5.3	11.7	16.9	19.0	20.4	19.1	15.4	11.4	7.3	2.0		
		MCDB	1.6	3.4	7.5	15.8	22.2	24.4	25.6	24.3	18.9	13.2	8.3	2.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.5	5.3	6.9	9.5	11.0	10.1	10.4	11.5	9.8	8.3	4.5	4.6		
		MCDBR	3.7	5.7	10.7	13.0	13.3	12.9	13.2	14.7	14.6	12.1	6.0	4.6		
	5% WB	MCWBR	3.2	4.7	6.6	6.5	6.2	6.3	5.8	6.3	7.1	7.5	4.7	4.2		
		MCWBR	3.4	5.3	9.7	11.0	11.7	11.7	11.8	12.7	12.3	11.0	5.5	4.5		
(40) Clear-Sky Solar Irradiance	taub	taub	0.309	0.341	0.372	0.438	0.409	0.397	0.429	0.431	0.399	0.363	0.344	0.313		
		taud	2.171	2.103	2.166	2.103	2.267	2.322	2.246	2.225	2.276	2.347	2.274	2.216		
	Ebn at Noon	Ebn at Noon	682	758	814	798	847	860	822	798	780	740	641	615		
		Edh at Noon	75	105	120	143	127	122	129	126	107	83	69	62		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.85	1.70	3.00	4.29	5.66	5.99	5.78	5.04	3.40	1.88	0.76	0.55		
		RadStd	0.16	0.28	0.47	0.41	0.46	0.49	0.38	0.36	0.45	0.31	0.12	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		+0.70	N/A	N/A	+1.01	+0.44	N/A	-131	-189	+110	+52

Nomenclature: See separate page